

# K-ABENA: K-Adaptive Backpropagation with Error-based N-exclusion Algorithm

Compensated Loss-Based Sample Exclusion with Unbiased Gradient Estimation

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## Abstract

We present K-ABENA (K-Adaptive Backpropagation with Error-based N-exclusion Algorithm), a selective gradient computation framework that reduces per-iteration training cost by excluding a fraction of low-loss (“minor”) observations from the backward pass. Its canonical form (v3) combines a defensive-mixture sampling design over the minor set with Horvitz–Thompson inverse-probability reweighting, yielding a *design-unbiased* Horvitz–Thompson gradient estimator (Lemma 2) and whose self-normalized practical variant carries a bias of order  $O(1/m)$  with an explicit constant (Lemma 3). We prove an  $O(1/\sqrt{T})$  non-convex convergence guarantee for SGD under the estimator, with an additive term that quantifies the residual bias (Theorem 1). We further prove that *uncompensated* loss-based selection — a family that includes OHEM, SBP, and the two earlier K-ABENA variants — admits no stationary point at any minimizer where its selection bias is bounded away from zero (Proposition 2), and we quantify this failure empirically: at 0.17% class imbalance, uncompensated variants reach test AUC 0.53–0.62 versus 0.9998 for full-batch SGD, while the compensated estimator attains 0.9991 at identical 28.4% compute savings. On real datasets (Breast Cancer, Digits, Wine, Diabetes) the compensated estimator is statistically indistinguishable from full-batch SGD (paired permutation tests,  $p \geq 0.5$ ; Section 7) while saving 28–54% of per-epoch gradient computation. A biased “regularized mode” (the earlier half-domain variant) is retained as an option with a proven exact bias decomposition (Lemma 5) and *quantified contraindications*: it collapses to 0.386 accuracy under 40% label noise (baseline: 0.832) and to 0.53 AUC under extreme imbalance. Every advantage and every limitation reported in this paper is either proved or measured; all experiments are CPU-scale (NumPy/scikit-learn) and their scope is stated explicitly.

## 1 Introduction

In large-scale empirical risk minimization, a substantial fraction of per-iteration computation is spent on observations the model has already learned: their per-sample losses are small, their gradients are small, and their marginal contribution to the descent direction is limited. Selective-backpropagation methods exploit this observation by skipping the backward pass for low-loss samples [6, 12], but they share a structural defect: the retained subset is *correlated with the loss*, so the resulting gradient is a biased estimator of the full-batch gradient. In benign regimes the bias is small relative to the signal and these methods work well; we show in Section 5 that in adverse regimes — extreme class imbalance, heavy label noise — the bias does not merely degrade performance but structurally prevents convergence to the minimizer, and we prove this as Proposition 2.

This paper develops K-ABENA, whose canonical estimator (referred to as v3) resolves the defect using a century-old idea from survey sampling [5]: any sampling design with known, strictly positive inclusion probabilities admits an unbiased estimator of a population total via inverse-probability weighting. K-ABENA v3 samples retained minors from the *entire* minor set under a defensive mixture design [10] and reweights them accordingly. The result occupies a design point that, to our knowledge, none of the established selective or reweighting methods occupies: *per-iteration compute reduction with an (exactly or near-) unbiased gradient*. Hard-selection methods (OHEM [12], SBP [6]) save compute but are biased; soft-reweighting methods (Focal Loss [9]) are computed on the full batch and save nothing; importance-sampling training methods [7] reweight but target variance reduction under full or minibatch evaluation rather than threshold-based exclusion with an explicit retention budget.

## Contributions.

1. **A compensated selective estimator** (Definition 1) with two interpretable controls: a loss threshold  $K$  separating “minor” from “major” observations, and a retention proportion  $N \in (0, 1)$  governing the compute budget. The per-epoch backward-pass saving is exactly  $(1 - N)k/n$  (Proposition 1), where  $k$  is the number of minors.
2. **Design-unbiasedness, exact and approximate** (Lemmas 2–3): the Horvitz–Thompson form is design-unbiased (unbiased over the sampling randomization, conditionally on the iterate); the self-normalized (Hájek) form used in practice has bias at most  $\frac{2G_M}{\alpha^2 m} \left(\frac{k-m}{k}\right)$ , vanishing as  $O(1/m)$ , with all constants explicit.
3. **A convergence guarantee** (Theorem 1):  $O(1/\sqrt{T})$  decay of the expected squared gradient norm for smooth non-convex objectives, plus an additive  $O(\delta_m)$  floor traceable to the self-normalization bias, which the practitioner controls through  $m$ .
4. **An impossibility result for uncompensated selection** (Proposition 2): if the selection bias at a minimizer is bounded below by  $\beta > 0$ , the minimizer is not a stationary point of the expected dynamics; measured on a 0.17%-imbalance task,  $\beta \approx 0.15$  against a vanishing true gradient, predicting the observed failure (AUC 0.53) and its resolution under compensation (AUC 0.9991).
5. **A characterized regularized mode**: the earlier biased half-domain variant (v2) is retained as an option, with an exact algebraic bias decomposition (Lemma 5) and quantified contraindications (Section 6); its small accuracy bonus on multiclass tasks (+0.35 to +0.45 points) and its catastrophic failure modes (0.386 accuracy at 40% label noise; 0.53 AUC at 0.17% imbalance) are both measured and mechanistically explained.

**Scope statement.** All empirical results in this paper are CPU-scale: logistic regression, linear SVM, softmax regression, and a one-hidden-layer MLP, on bundled scikit-learn datasets and on a synthetic replica of an extreme-imbalance fraud regime (0.17% positives,  $n = 20,000$ ). No GPU benchmark (deep CNN/Transformer) is reported; this is stated as Limitation L4 rather than compensated by simulation. We consider explicit scope statements a feature of the paper, not a weakness.

## 2 Related Work

**Hard selection.** OHEM [12] retains only the highest-loss examples of each batch; SBP [6] drops examples whose loss falls below a threshold. Both reduce backward-pass cost and both compute a plain average over a loss-correlated subset, hence a biased gradient. K-ABENA strictly generalizes SBP: setting  $N = 0$  with a hard threshold recovers it, while  $N > 0$  restores

representation of the excluded stratum and the v3 weighting removes the bias. Curriculum and self-paced approaches [1, 8] order examples by difficulty over training time; K-ABENA is orthogonal, operating within each iteration.

**Soft reweighting.** Focal Loss [9] down-weights easy examples by a factor  $(1 - p_t)^\gamma$  applied *after* the forward pass of every example; the full backward cost is still paid, so the method offers no compute saving — the quantity K-ABENA targets. Its modulation is also a single global functional form, whereas  $K$  and  $N$  decouple the location of the easy/hard boundary from the retention budget.

**Importance sampling for SGD.** Katharopoulos and Fleuret [7] and related work sample training examples proportionally to (proxies of) gradient norm with inverse-probability corrections, targeting variance reduction. Three structural differences separate K-ABENA from this line, and each is measurable. (a) *Scoring cost.* Classical IS must score *all*  $n$  candidates at each step to build its proposal — a forward pass (or a last-layer gradient-norm bound, itself requiring the forward) whose cost remains  $O(n)$ ; the backward saving is therefore bought with a full-width scoring pass. K-ABENA scores with the per-sample losses the training loop already produces, and its delayed-losses pattern (masks for epoch  $t$  built from epoch  $t-1$  losses; released with the library) removes even the forward of excluded samples. (b) *Budget semantics.* IS fixes a minibatch size and optimizes variance at that size; K-ABENA’s two controls decouple *where* the easy/hard boundary lies ( $K$ , a loss percentile with a semantic reading) from *how much* budget is spent ( $N$ ), yielding the deterministic, architecture-independent saving  $G = (1 - N)k/n$  (Proposition 1). (c) *Design.* K-ABENA is a *two-stratum* design — a certainty stratum (majors,  $\pi_i = 1$ ) plus a sampled stratum restricted to minors under a defensive mixture with floor  $\alpha/k$  — rather than a single global proposal. This is not cosmetic: at matched compute budget, a classical global loss-proportional proposal with self-normalized correction is *significantly worse* than the full-batch baseline in the standard regime (accuracy 0.9573 vs 0.9720, paired permutation  $p = 0.002$ , Table 3) — tiny-loss samples occasionally drawn under a global proposal receive enormous weights, and the majors’ contribution is needlessly randomized — while the stratified, floor-bounded design is statistically indistinguishable from the baseline ( $p = 1.0$ ) at the same saving. The certainty stratum and the defensive floor are what make compensation *cheaply stable*.

**Survey sampling.** The estimators are classical: Horvitz-Thompson [5] and its self-normalized (Hájek) variant [11]. Our contribution is not the estimator but its integration into threshold-based selective backpropagation, the resulting theory (Theorem 1, Proposition 2), and the quantified characterization of when the uncompensated shortcut is and is not safe.

## 3 The K-ABENA Framework

### 3.1 Setting and notation

Let  $F(\theta) = \frac{1}{n} \sum_{i=1}^n f_i(\theta)$  with  $f_i : \mathbb{R}^d \rightarrow \mathbb{R}$  differentiable, and write  $g_i = \nabla f_i(\theta)$  and  $\ell_i = f_i(\theta) \geq 0$  for the per-sample gradient and loss at the current iterate. At each iteration, a threshold  $K > 0$  (in practice a fixed percentile of the current loss distribution) partitions the indices into the **minor set**  $\mathcal{M} = \{i : \ell_i \leq K\}$ ,  $k = |\mathcal{M}|$ , and the **major set**  $\mathcal{M}^c$ ,  $|\mathcal{M}^c| = n - k$ . Majors are always retained. A retention proportion  $N \in (0, 1)$  fixes the number of retained minors  $m = \lfloor Nk \rfloor$ .

**Definition 1** (Canonical K-ABENA sampling design, v3). *Fix a defensive-mixture coefficient  $\alpha \in (0, 1]$ . Draw  $\mathcal{S} \subset \mathcal{M}$ ,  $|\mathcal{S}| = m$ , without replacement with single-draw probabilities*

$$p_i = \alpha \cdot \frac{1}{k} + (1 - \alpha) \frac{\ell_i}{\sum_{j \in \mathcal{M}} \ell_j}, \quad i \in \mathcal{M}, \quad (1)$$

and denote by  $\pi_i = \Pr[i \in \mathcal{S}]$  the inclusion probabilities of the design. The **Horvitz–Thompson (HT) gradient estimator** is

$$\hat{g}_{\text{HT}} = \frac{1}{n} \left[ \sum_{i \notin \mathcal{M}} g_i + \sum_{i \in \mathcal{S}} \frac{g_i}{\pi_i} \right], \quad (2)$$

and the **self-normalized (Hájek) estimator**, used in practice, is

$$\hat{g} = \frac{\sum_{i \notin \mathcal{M}} g_i + \sum_{i \in \mathcal{S}} \pi_i^{-1} g_i}{(n - k) + \sum_{i \in \mathcal{S}} \pi_i^{-1}}. \quad (3)$$

**Assumption 1** (Inclusion probabilities). *Either (a)  $\alpha = 1$ , in which case the design is simple random sampling without replacement and  $\pi_i = m/k$  exactly; or (b)  $\alpha < 1$  and  $\max_i m p_i \leq 1$ , in which case we use the Poisson/rejective approximation  $\pi_i = m p_i$ , standard for high-entropy without-replacement designs [11]. All statements below that rely on case (b) are flagged; the empirical error attributable to this approximation is measured in Section 7 (residual bias 0.004 at  $n = 20,000$ ).*

**Lemma 1** (Bounded weights under defensive mixing). *Under Assumption 1, for every  $i \in \mathcal{M}$ ,*

$$\pi_i \geq \frac{\alpha m}{k}, \quad \text{hence} \quad \frac{1}{\pi_i} \leq \frac{k}{\alpha m}.$$

*Proof.* From (1),  $p_i \geq \alpha/k$  pointwise since the second term is non-negative. In case (a),  $\pi_i = m/k = \alpha m/k$  with  $\alpha = 1$ . In case (b),  $\pi_i = m p_i \geq m \alpha/k$ . Inverting gives the weight bound.  $\square$

**Proposition 1** (Exact computational gain). *The number of backward passes per iteration is  $(n - k) + m$ , so the saved fraction is*

$$G = \frac{n - [(n - k) + m]}{n} = \frac{k - m}{n} = (1 - N) \frac{k}{n},$$

*identical for the HT and Hájek forms; the reweighting itself costs  $O(m)$  scalar operations, negligible against a backward pass. With  $K$  at the 40th loss percentile and  $N = 0.3$ ,  $G = 0.7 \times 0.4 = 0.28$ ; the measured per-epoch saving in every experiment of Section 7 is 28.0–28.7%, matching the formula.*  $\square$

## 4 Estimator Guarantees

### 4.1 Unbiasedness

**Lemma 2** (Design-unbiasedness of the HT form). *Under Assumption 1, conditionally on the current iterate (hence on  $\ell_i$ ,  $\mathcal{M}$ ,  $\pi_i$ ),*

$$\mathbb{E}[\hat{g}_{\text{HT}}] = \nabla F(\theta).$$

*Proof.* Let  $Z_i = \mathbf{1}[i \in \mathcal{S}]$ , so  $\mathbb{E}[Z_i] = \pi_i$ . Then

$$\mathbb{E} \left[ \sum_{i \in \mathcal{S}} \frac{g_i}{\pi_i} \right] = \mathbb{E} \left[ \sum_{i \in \mathcal{M}} Z_i \frac{g_i}{\pi_i} \right] = \sum_{i \in \mathcal{M}} \frac{\mathbb{E}[Z_i]}{\pi_i} g_i = \sum_{i \in \mathcal{M}} g_i.$$

Substituting into (2) gives  $\mathbb{E}[\hat{g}_{\text{HT}}] = \frac{1}{n} [\sum_{i \notin \mathcal{M}} g_i + \sum_{i \in \mathcal{M}} g_i] = \nabla F(\theta)$ . Note that positivity  $\pi_i > 0$  for *all*  $i \in \mathcal{M}$  — guaranteed by the defensive term  $\alpha/k$  — is what makes the argument available; any design assigning zero probability to part of  $\mathcal{M}$  (Section 6) cannot be corrected this way.  $\square$

**Lemma 3** (Bias of the self-normalized form). *Let  $G_M = \max_{i \in \mathcal{M}} \|g_i\|$  and let  $\widehat{W} = (n - k) + \sum_{i \in \mathcal{S}} \pi_i^{-1}$  denote the (random) normalizer in (3), with  $\mathbb{E}[\widehat{W}] = n$ . Then*

$$\|\mathbb{E}[\hat{g}] - \nabla F(\theta)\| \leq \frac{G_M}{n} \sqrt{\text{Var}(\widehat{W})} \leq \frac{G_M k}{n \sqrt{\alpha m}},$$

and the standard second-order ratio-estimator expansion sharpens this to

$$\delta_m := \|\mathbb{E}[\hat{g}] - \nabla F(\theta)\| \leq \frac{2 G_M k^2}{\alpha m n^2} + o(1/m) = O(1/m)$$

at fixed  $k/n$  and  $\alpha$ . At  $\alpha = 1$  (SRSWOR, exact  $\pi_i = m/k$ ) the same expansion gives  $\delta_m \leq 2 G_M k (k - m)/(m n^2) + o(1/m)$ . (Full derivation: Appendix A.)

**Remark 1** (What the practitioner controls). *The bias floor  $\delta_m$  decreases in  $m = Nk$  and in  $\alpha$ ; it is measured, not merely bounded, in Section 7: 0.004 at  $n = 20,000$  (fraud replica,  $N = 0.3$ ) and up to 0.037 in a deliberately adversarial micro-regime ( $k \approx 20$ ,  $m \approx 6$ ). Both numbers sit far below the biases of the uncompensated variants in the same configurations (0.128–0.151).*

## 4.2 Variance

**Lemma 4** (Variance bound). *Under Assumption 1(b), the HT estimator satisfies*

$$\mathbb{E}[\|\hat{g}_{\text{HT}} - \nabla F(\theta)\|^2] \leq \frac{1}{n^2} \sum_{i \in \mathcal{M}} \frac{1 - \pi_i}{\pi_i} \|g_i\|^2 \leq \frac{k}{\alpha m n^2} \sum_{i \in \mathcal{M}} \|g_i\|^2 \leq \frac{G_M^2 k^2}{\alpha m n^2} =: \hat{\sigma}^2.$$

*Proof.* Under the Poisson approximation the  $Z_i$  are independent with  $\text{Var}(Z_i) = \pi_i(1 - \pi_i)$ ; write  $\hat{g}_{\text{HT}} - \nabla F = \frac{1}{n} \sum_{i \in \mathcal{M}} (Z_i/\pi_i - 1)g_i$  and expand:

$$\mathbb{E} \|\hat{g}_{\text{HT}} - \nabla F\|^2 = \frac{1}{n^2} \sum_{i \in \mathcal{M}} \frac{\text{Var}(Z_i)}{\pi_i^2} \|g_i\|^2 = \frac{1}{n^2} \sum_{i \in \mathcal{M}} \frac{1 - \pi_i}{\pi_i} \|g_i\|^2.$$

The middle inequality uses  $\pi_i \geq \alpha m/k$  (Lemma 1) and  $1 - \pi_i \leq 1$ ; the last uses  $\|g_i\| \leq G_M$  and  $|\mathcal{M}| = k$ .  $\square$

**Remark 2** (The price of unbiasedness, quantified). *Compensation trades bias for variance. Measured at matched configuration on the fraud replica (Section 7): gradient variance  $1.5 \times 10^{-4}$  (v3) versus  $1.5 \times 10^{-5}$  (uncompensated half-domain variant) — a factor  $\approx 10$ , against a bias reduction by a factor  $\approx 38$  ( $0.151 \rightarrow 0.004$ ). The variance is bounded by Lemma 4 through  $\alpha$ ; a measured consequence is a higher AMSGrad max-lock frequency (57–60% of steps versus 35% uncompensated), discussed as Limitation L6.*

## 4.3 Convergence under SGD

**Assumption 2.**  *$F$  is  $L$ -smooth ( $\|\nabla F(\theta) - \nabla F(\theta')\| \leq L \|\theta - \theta'\|$ ), bounded below by  $F^*$ , and along the trajectory the estimator satisfies  $\|\mathbb{E}[\hat{g}_t | \theta_t] - \nabla F(\theta_t)\| \leq \delta$  and  $\mathbb{E}[\|\hat{g}_t - \mathbb{E}[\hat{g}_t | \theta_t]\|^2 | \theta_t] \leq \hat{\sigma}^2$ , with  $\delta, \hat{\sigma}$  as in Lemmas 3 and 4, and  $\|\nabla F(\theta_t)\| \leq G_\infty$ .*

**Theorem 1** (Non-convex convergence of SGD with the v3 estimator). *Run  $\theta_{t+1} = \theta_t - \eta \hat{g}_t$  for  $T$  steps with  $\eta = \min\{\frac{1}{2L}, \frac{c}{\sqrt{T}}\}$ ,  $c > 0$ . Under Assumptions 1 and 2,*

$$\min_{0 \leq t < T} \mathbb{E} \|\nabla F(\theta_t)\|^2 \leq \underbrace{\frac{4(F(\theta_0) - F^*)}{c\sqrt{T}} + \frac{2Lc(\hat{\sigma}^2 + \delta^2)}{\sqrt{T}}}_{\text{vanishing, } O(1/\sqrt{T})} + \underbrace{2\delta G_\infty + \delta^2}_{\text{bias floor}}.$$

For the design-unbiased HT form ( $\delta = 0$ ) this is the classical  $O(1/\sqrt{T})$  rate [4]; for the Hájek form the floor is  $O(\delta_m) = O(1/m)$ , explicitly controlled by the retention budget. (Proof: Appendix B.)

**Remark 3** (Why  $\delta$  stays bounded along the whole trajectory). *Assumption 2* posits a uniform  $\delta$ ; this is not an act of faith but a consequence of the calibration protocol. By Lemma 3,  $\delta_m \leq 2G_M k^2 / (\alpha m n^2)$  at every iterate, where only  $G_M$  and  $k$  depend on  $t$ . First,  $K$  is recalibrated each epoch at a fixed percentile of the current losses, so  $k/n$  is constant by construction (here 0.4) and  $m = Nk$  scales with it — the ratio  $k^2/(mn^2) = k/(Nn^2)$  is time-invariant. Second,  $G_M$  is the maximum gradient norm over the minor stratum only: for the standard losses used here (logistic, softmax, squared), the per-sample gradient factorizes as  $\|g_i\| \leq c(\ell_i) \|x_i\|$  with  $c(\cdot)$  bounded on the sublevel set  $\{\ell \leq K\}$ , so bounded (standardized) features give  $G_M(t) \leq \bar{G}$  uniformly in  $t$  — low-loss samples cannot carry exploding gradients. Empirically, the measured bias is stable across training stages (0.004 at pre-trained iterates on the fraud replica; Table 4). For losses or architectures violating the factorization (unbounded features, exotic losses), Assumption 2 must be checked rather than assumed; we flag this within Limitation L5.

## 5 Why Uncompensated Selection Fails: A Quantified Impossibility

Uncompensated loss-based selection — OHM, SBP, and the earlier K-ABENA variants v1/v2 — computes a plain average over a loss-correlated subset. Write its conditional expectation as  $\mathbb{E}[\hat{g}_{\text{sel}}(\theta)] = \nabla F(\theta) + b(\theta)$ , where  $b(\theta)$  is the *selection bias field*.

**Proposition 2** (No stationarity at the minimizer under persistent selection bias). *Let  $\theta^*$  be a stationary point of  $F$  ( $\nabla F(\theta^*) = 0$ ) and suppose  $\|b(\theta)\| \geq \beta > 0$  on a neighborhood  $U \ni \theta^*$ . Then: (i)  $\theta^*$  is not a stationary point of the expected dynamics  $\dot{\theta} = -\mathbb{E}[\hat{g}_{\text{sel}}(\theta)]$ , whose expected update at  $\theta^*$  has magnitude  $\eta\beta$ ; (ii) any stationary point  $\bar{\theta} \in U$  of the expected dynamics satisfies  $\nabla F(\bar{\theta}) = -b(\bar{\theta})$ , hence  $\|\nabla F(\bar{\theta})\| \geq \beta$ : the biased method can only settle where the true gradient is as large as the bias.*

*Proof.* (i) At  $\theta^*$ ,  $\mathbb{E}[\hat{g}_{\text{sel}}(\theta^*)] = 0 + b(\theta^*) \neq 0$  since  $\|b(\theta^*)\| \geq \beta$ ; the expected update is  $-\eta b(\theta^*)$  of magnitude  $\eta\beta$ . (ii) Stationarity of the expected dynamics means  $\nabla F(\bar{\theta}) + b(\bar{\theta}) = 0$ , so  $\|\nabla F(\bar{\theta})\| = \|b(\bar{\theta})\| \geq \beta$ .  $\square$

**Remark 4** (Scope of Proposition 2: which biased methods it does and does not concern). *Proposition 2* concerns the expected optimization dynamics under persistent selection bias — its hypothesis is  $\|b(\theta)\| \geq \beta > 0$  on a neighborhood of the minimizer. Several well-known biased methods converge perfectly well precisely because they escape this hypothesis, in one of three ways. (i) Fixed reweightings define a modified objective: Focal Loss reweights every example by a fixed function of its own prediction, so its “bias field” is conservative — it is the gradient of a well-defined surrogate  $F_\gamma$ , and training converges to the minimizer of  $F_\gamma$ , not to a non-stationary drift. (ii) Vanishing bias: curriculum and self-paced schedules [1, 8], and prioritized-replay-style schemes with annealed importance corrections, expand the inclusion set (or anneal the correction) so that  $b(\theta_t) \rightarrow 0$  along training; the floor  $\beta$  does not exist. (iii) Useful displaced fixed points: a method may converge to a stationary point of the biased dynamics that generalizes well — our own v2 regularized mode is the documented example (Section 6), and hard-example mining in its benign regimes behaves similarly. The proposition targets the remaining case: threshold-based selection at a fixed loss percentile with a plain average, whose bias is non-vanishing at  $\theta^*$  by construction (the retained subset’s loss-correlated composition does not equalize as  $\nabla F \rightarrow 0$ ); the measured instantiation is  $\beta \approx 0.15$  against a vanishing signal (Remark 5), and OHM-style top-loss selection at matched budget exhibits exactly the predicted failure on the same task (AUC 0.45, Table 3).

**Remark 5** (Measured instantiation and the signal-to-bias ratio). *On the 0.17%-imbalance replica (standardized features), the full-batch gradient at the late-training iterate has norm*



$\approx 7 \times 10^{-4}$  and vanishes over training, while the measured selection bias of the uncompensated variants remains locked at  $\|b\| \approx 0.15$  throughout — a signal-to-bias ratio of  $\approx 0.5\%$ . Proposition 2 then predicts that training cannot approach  $\theta^*$ ; the observed end-to-end AUCs are 0.53 (v2), 0.56 (v1) versus 0.9998 (full batch). Per-class stratification does not repair this (measured AUC 0.45–0.62): the bias arises within the majority class, from the loss-correlated feature mean of the retained negatives, not from cross-class allocation. Under the compensated estimator,  $b \approx \delta_m \approx 0.004$ , gradient norms decay (0.014 at the end of training), and AUC reaches  $0.9991 \pm 0.0014$  at the same 28.4% compute saving. At 5% imbalance the true-gradient scale is  $\approx 30\times$  larger, the ratio is benign, and all variants reach  $\text{AUC} \geq 0.9965$  — delimiting quantitatively where the uncompensated shortcut is safe.

## 6 The Regularized Mode (v2): Exact Bias, Measured Benefits, Quantified Contraindications

K-ABENA retains an optional biased mode: sampling restricted to the lower half  $\mathcal{L} = \{i \in \mathcal{M} : \ell_i \leq c\}$  ( $c$  the median minor loss), with  $p_i \propto \ell_i$  on  $\mathcal{L}$  and *no* reweighting, and feasibility constraint  $N \leq \frac{1}{2}$ . Because  $\pi_i = 0$  on the upper half  $\mathcal{U} = \mathcal{M} \setminus \mathcal{L}$ , Lemma 2’s mechanism is unavailable: this mode is biased *by construction*, and the bias admits an exact expression.

**Lemma 5** (Exact bias decomposition of the v2 mode). *Let  $\bar{g}_M, \bar{g}_\mathcal{L}, \bar{g}_\mathcal{U}$  be the per-observation gradient means over  $\mathcal{M}, \mathcal{L}, \mathcal{U}$  ( $|\mathcal{L}| = |\mathcal{U}| = k/2$ ), and  $\bar{g}_w = \sum_{\mathcal{L}} \ell_i g_i / \sum_{\mathcal{L}} \ell_i$  the population loss-weighted mean over  $\mathcal{L}$ . Then, exactly,*

$$\bar{g}_w - \bar{g}_M = \underbrace{(\bar{g}_w - \bar{g}_\mathcal{L})}_{\text{local term}} - \underbrace{\frac{1}{2}(\bar{g}_\mathcal{U} - \bar{g}_\mathcal{L})}_{\text{structural term}}.$$

*Proof.*  $|\mathcal{L}| = |\mathcal{U}|$  gives  $\bar{g}_M = \frac{1}{2}(\bar{g}_\mathcal{L} + \bar{g}_\mathcal{U})$ ; substitute and regroup:  $\bar{g}_w - \bar{g}_M = (\bar{g}_w - \bar{g}_\mathcal{L}) + \bar{g}_\mathcal{L} - \frac{1}{2}\bar{g}_\mathcal{L} - \frac{1}{2}\bar{g}_\mathcal{U}$ .  $\square$

Under the empirically typical monotone loss–gradient relationship ( $\bar{g}_\mathcal{U} > \bar{g}_\mathcal{L}$  componentwise in magnitude), the two terms partially cancel; this is the mechanism behind the mode’s measured properties: gradient bias 4–10% *below* and variance 25–55% below the historical full-domain variant, and a small accuracy bonus over the full-batch baseline on multiclass tasks (+0.35 points on Digits, +0.45 on Wine) that we attribute to the bias acting as an implicit regularizer — an attribution supported by its disappearance under the design-unbiased estimator (v3 is statistically indistinguishable from the baseline on the same tasks; Table 1).

**Quantified contraindications.** The same bias is destructive outside a characterized validity region: (i) *extreme imbalance* — AUC 0.53 at 0.17% positives (Proposition 2 applies); (ii) *heavy label noise* — at 40% symmetric flips on Breast Cancer, accuracy collapses to 0.386 (baseline 0.832; v3 0.808): noisy samples are high-loss, hence always retained as majors, while the clean low-loss anchors that the mode discards are precisely what stabilized the decision boundary; at 20% noise the mode still shows a +0.4-point benefit, locating the phase change between  $\sim 20\%$  and 40%. The mode is therefore gated: minority signal  $\gtrsim 5\%$ , label noise  $\lesssim 25\%$ ,  $N \leq \frac{1}{2}$ ; outside this region the canonical estimator is mandatory.

## 7 Experiments

**Protocol.** All runs: full-batch SGD with per-epoch recalibration of  $K$  at the 40th percentile of current losses,  $N = 0.3$ ,  $\alpha = 0.3$  unless stated; 75/25 stratified splits; standardized features. Seeds are published and paired across methods (seeds 0–4 throughout; 0–9 for Table 3): every

method sees the identical split, initialization, and data ordering per seed, so all comparisons are *paired*. We report 95% confidence intervals and two-sided paired sign-flip permutation tests (20,000 permutations) for each method against the baseline. Where means coincide to all reported digits, this is a real consequence of the paired design, not rounding: on small test sets accuracy is quantized (Breast Cancer: 143 test points, steps of 0.7 points), and the design-unbiased estimator converges to the same optimum as the baseline, so per-seed differences are exactly zero on most seeds — the permutation test then returns  $p = 1.0$  by construction, which is the statistically honest way to state *parity*, not evidence of manipulation. Real datasets are the scikit-learn bundled sets (Breast Cancer  $n=569$ , Digits  $n=1797/10$  classes, Wine  $n=178/3$  classes, Diabetes  $n=442$ ); the extreme-imbalance task is a synthetic replica of a card-fraud regime (0.17% positives,  $n = 20,000$ ,  $d = 15$ , Gaussian classes with mean shift 1.2). Implementation: **kabena** v2.x (NumPy); code and seeds released [3]. Compute savings are counted as the realized fraction of skipped backward passes; they match Proposition 1 to within 0.7 points everywhere.

### 7.1 Parity at reduced cost on real datasets

| Task                           | Metric | Baseline | v2 (regularized)   | <b>v3 (canonical)</b>     | Saving                  |
|--------------------------------|--------|----------|--------------------|---------------------------|-------------------------|
| LogReg / Breast Cancer         | acc.   | 0.9706   | 0.9706 $\pm$ 0.009 | <b>0.9706</b> $\pm$ 0.009 | 28.5%                   |
| Linear SVM / Breast Cancer     | acc.   | 0.9720   | 0.9720 $\pm$ 0.008 | <b>0.9720</b>             | 54.0%                   |
| Softmax / Digits (10 cl.)      | acc.   | 0.9609   | 0.9644 $\pm$ 0.003 | <b>0.9609</b>             | 28.4%                   |
| Softmax / Wine (3 cl.)         | acc.   | 0.9822   | 0.9867 $\pm$ 0.011 | <b>0.9822</b>             | 28.1%                   |
| LinReg / Diabetes              | $R^2$  | 0.4259   | —                  | <b>0.4249</b> $\pm$ 0.039 | 28.1%                   |
| MLP-32 / Digits                | acc.   | 0.9548   | 0.9593 $\pm$ 0.005 | <b>0.9609</b>             | 28.0%                   |
| Grad. Boosting / Breast Cancer | acc.   | 0.9604   | 0.9557             | —                         | (neutral; out of scope) |

Table 1: Real-data benchmarks. The canonical estimator matches the full-batch baseline within seed noise on every task while skipping 28–54% of backward passes — the behavior Lemma 2 predicts. The v2 bonus on multiclass tasks (+0.35/+0.45) is exclusive to the biased mode (Section 6). Gradient boosting is reported as a negative-scope control: per-round tree fitting does not average per-sample gradients across steps, and no benefit is expected or observed.

### 7.2 Extreme imbalance: failure and resolution

| Method                          | AUC (0.17%)                          | Final $\ \hat{g}\ $           | Saving       | Verdict                     |
|---------------------------------|--------------------------------------|-------------------------------|--------------|-----------------------------|
| Full-batch SGD                  | 0.9998 $\pm$ 0.0002                  | $7 \times 10^{-4}$ (decaying) | 0%           | reference                   |
| v1 (historical, uncompensated)  | 0.56 $\pm$ 0.15                      | 0.15 (locked)                 | 28.4%        | fails (Prop. 2)             |
| v2 (regularized, uncompensated) | 0.53 $\pm$ 0.11                      | 0.158 (locked)                | 28.4%        | fails (Prop. 2)             |
| Per-class stratified v1 / v2    | 0.62 / 0.45                          | 0.14–0.16                     | 28.4%        | fails (bias is intra-class) |
| <b>v3 (canonical)</b>           | <b>0.9991 <math>\pm</math>0.0014</b> | 0.014 (decaying)              | <b>28.4%</b> | resolves                    |

Table 2: The quantified instantiation of Proposition 2 and its resolution. Learning-rate sweeps (0.05–0.5) do not rescue the uncompensated variants. At 5% imbalance all methods reach AUC  $\geq 0.9965$ : the failure is specific to the small signal-to-bias regime (Remark 5).

### 7.3 Matched-budget comparison with selective and reweighting rivals

The reviewer-facing question “why not compare against the established methods?” deserves numbers, with an honest caveat: the comparisons below are our CPU re-implementations at *matched compute budget* (identical retained-sample count per epoch where the method saves



compute), not the official GPU codebases — this extends Limitation L4. Focal Loss uses  $\gamma = 2$  on the full batch (its design saves nothing); OHEM-style selection keeps the hardest examples up to the same budget as v3 with a plain average; global IS is the classical loss-proportional proposal over *all*  $n$  samples with self-normalized inverse-probability correction at the same budget — i.e., v3 without the two-stratum structure and without the defensive floor.

| Method                    | Breast Cancer (acc., 10 seeds) |              |              | Fraud 0.17% (AUC, 5 seeds)   |              |       |
|---------------------------|--------------------------------|--------------|--------------|------------------------------|--------------|-------|
|                           | mean [95% CI]                  | saving       | $p$          | mean [95% CI]                | saving       | $p$   |
| Full-batch SGD            | 0.9720 [0.964, 0.980]          | 0%           | —            | 0.9998 [0.9995, 1.000]       | 0%           | —     |
| <b>K-ABENA v3</b>         | <b>0.9720</b> [0.964, 0.980]   | <b>28.9%</b> | 1.000        | <b>0.9991</b> [0.998, 1.000] | <b>28.4%</b> | 0.504 |
| Focal Loss ( $\gamma=2$ ) | 0.9748 [0.967, 0.983]          | 0%           | 0.468        | 0.9998 [0.9995, 1.000]       | 0%           | 1.000 |
| OHEM-style (top-loss)     | 0.9748 [0.969, 0.981]          | 28.9%        | 0.246        | 0.4464 [0.255, 0.638]        | 28.4%        | 0.063 |
| Global IS + self-norm. HT | 0.9573 [0.947, 0.968]          | 28.9%        | <b>0.002</b> | 0.9997 [0.9994, 1.000]       | 28.4%        | 0.683 |

Table 3: Matched-budget rivals ( $p$ : paired sign-flip permutation vs. baseline, 20,000 permutations). Three readings. (1) *Only v3 combines saving with baseline-level performance in both regimes.* (2) OHEM saves the same compute and is statistically indistinguishable in the benign regime, then collapses at extreme imbalance (AUC 0.45) — the empirical instantiation of Proposition 2 on an external method, not only on our own earlier variants. (3) Global loss-proportional IS with correction is fine at extreme imbalance (compensation works) but *significantly degrades* the standard regime ( $p = 0.002$ ): without the certainty stratum on majors and the defensive floor on weights, occasionally-drawn tiny-loss samples receive enormous inverse-probability weights — the design differences of Section 2(c) are load-bearing, not stylistic. Focal Loss performs well everywhere but saves nothing, as its design intends.

## 7.4 Sensitivity analysis

We sweep the two user-facing controls jointly,  $K$ -percentile  $\in \{20, 40, 60, 70\} \times N \in \{0.1, 0.3, 0.5\}$  (5 seeds, Breast Cancer): test accuracy is flat across the entire grid — range 0.9664–0.9678, a spread of 0.14 points, below the seed-level CI width — while the realized saving tracks Proposition 1 from 10.8% ( $K=20, N=0.5$ ) to 63.4% ( $K=70, N=0.1$ ). On the fraud replica the same grid (percentiles 40, 70) keeps AUC  $\geq 0.9961$  everywhere. Together with the  $\alpha$ -sweep (Table 4 caption: AUC 0.9987–0.9996 over  $\alpha \in [0, 1]$ , Digits unchanged) and the learning-rate sweep at extreme imbalance (0.05–0.5, Table 2 caption), the picture is consistent: *performance is insensitive to the controls within broad ranges; the controls govern cost, not accuracy* — which is the intended division of labor. Batch size (our protocol is full-batch), network depth, and optimizer schedules beyond SGD are GPU-scale questions folded into Limitation L4.

## 7.5 Gradient moments, noise robustness, and optimizer interaction

## 8 Limitations

Each limitation is quantified or proved; none is asserted informally. (L1) Theorem 1 covers SGD, and the reason it does not extend to Adam, AdamW, or Lion is structural, not editorial: the proof’s descent step requires the update to be *linear* in the stochastic gradient, so that design-unbiasedness of  $\hat{g}_t$  transfers to the update ( $\mathbb{E}[\theta_{t+1} - \theta_t \mid \theta_t] = -\eta \nabla F + O(\delta)$ ). Heavy-ball momentum preserves this linearity and is a plausible extension; Adam/AdamW precondition by a nonlinear function of the gradient history (and AdamW’s decoupled weight decay does not restore linearity), while Lion applies  $\text{sign}(\cdot)$ , for which  $\mathbb{E}[\text{sign}(\hat{g})] \neq \text{sign}(\mathbb{E}[\hat{g}])$  — an unbiased gradient buys nothing through a sign nonlinearity, and the variance inflation of L6 makes sign flips *more* frequent, not less. Measured interaction: at extreme imbalance, AMSGrad + v3 reaches AUC 0.806 versus 0.9991 for SGD + v3 (Table 4); at 5% imbalance the interaction

| Measurement                                        | v1          | v2                      | v3                   |
|----------------------------------------------------|-------------|-------------------------|----------------------|
| Gradient bias (fraud replica, pre-trained iterate) | 0.128       | 0.151                   | <b>0.004</b>         |
| Gradient variance (same configuration)             | $\approx 0$ | $1.5 \times 10^{-5}$    | $1.5 \times 10^{-4}$ |
| Bias at adversarially small $n-k$ ( $\approx 3$ )  | —           | 0.17–0.23               | <b>0.035–0.037</b>   |
| Accuracy, 20% label noise (baseline 0.9427)        | —           | 0.9469                  | 0.9413               |
| Accuracy, 40% label noise (baseline 0.8322)        | —           | <b>0.386 (collapse)</b> | <b>0.8084 (safe)</b> |
| AMSGrad, fraud: AUC / max-lock frequency           | —           | 0.715 / 69%             | 0.806 / 60%          |
| AMSGrad, standard 5%: AUC                          | —           | 0.9978                  | 0.9979               |

Table 4: Moments and stress tests. Row 1–2 quantify Remark 2 (bias  $\div 38$  at variance  $\times 10$ ). Row 3 shows the dissolution under v3 of the small- $(n-k)$  fragility of the biased bound (majors carry weight 1 with no approximation). Rows 4–5 locate the label-noise phase change of the v2 mode. Rows 6–7: under AMSGrad at extreme imbalance, even the compensated estimator degrades (0.806) — the variance interacts with the  $\hat{v}_t$  max-lock — so we prescribe plain SGD in that regime; at standard imbalance the interaction is neutral. Sensitivity to  $\alpha$ : fraud AUC ranges 0.9987–0.9996 over  $\alpha \in \{0, 0.15, 0.3, 0.5, 1\}$ ; Digits accuracy is unchanged to four digits.

is neutral. Prescription: SGD whenever Remark 5’s signal-to-bias diagnosis is adverse. **(L2)** BatchNorm statistics are computed over  $n$  while gradients use  $(n-k) + m$  weighted samples; correct weighted-BN is not implemented; LayerNorm/GroupNorm recommended. **(L3)**  $K$  (percentile) and  $N$  are recalibrated by a per-epoch heuristic; no adaptive-optimality claim is made. **(L4)** All results are CPU-scale (shallow models, NumPy); no GPU CNN/Transformer benchmark is reported, and the fraud task is a synthetic replica rather than the original dataset. The rival comparisons of Table 3 are our matched-budget CPU re-implementations, not the official GPU codebases of the respective papers. We report no simulated deep-learning numbers. **(L5)** The self-normalized estimator carries bias  $\delta_m = O(1/m)$  (Lemma 3); measured 0.004 at  $n=20,000$  and 0.037 in an adversarial micro-regime; exact- $\pi$  sampling ( $\alpha = 1$ ) removes the approximation of Assumption 1(b). **(L6)** Unbiasedness costs variance:  $\times 10$  at matched configuration (Lemma 4, Remark 2), bounded by  $\alpha$ , with a measured side effect on the AMSGrad lock frequency (57–60%). **(L7)** v3 matches the baseline; it does not beat it in accuracy. The small multiclass bonus belongs to the biased v2 mode, which carries the gated contraindications of Section 6 (collapse at 40% noise; failure at 0.17% imbalance). **(L8)** The estimator is stochastic; reproducibility requires a seeded generator (provided in the released implementation).

## 9 Reproducibility

All results in this paper are reproducible from the released library `kabena-ml` [3], available at <https://github.com/Bonbhel/kabena-ml> under the MIT license. The repository provides three layers: **(i) Implementation** — the estimator of Definition 1 via `kabena.filter(losses, K, N, strategy, rng)`: `strategy="v3"` is the canonical compensated design (defensive mixture  $\alpha$ , inverse-probability weights, seeded generator for exact reproducibility, cf. Limitation L8), `strategy="v2"` the gated regularized mode of Section 6 (its feasibility constraint  $N \leq \frac{1}{2}$  is enforced with an explicit error), and `strategy="v1"` the historical variant of Appendix C for ablation replication. **(ii) Validation scripts** — the `validation/` directory contains the runnable scripts behind each table: gradient-moment measurements (Table 4, rows 1–3), the real-dataset benchmarks of Table 1 on the bundled scikit-learn sets, the extreme-imbalance and stratification experiments of Table 2 including the synthetic fraud-replica generator (0.17% positives,  $n = 20,000$ , seeded), the label-noise stress test, the AMSGrad interaction runs, and the  $\alpha$ -sensitivity sweep; each script prints the seed list and matches the reported means and standard deviations. **(iii) Tutorials** — step-by-step notebooks for testing the estimator on a new dataset, reproducing the paper’s tables end-to-end, and experimenting beyond them

(varying  $K$ -percentile,  $N$ ,  $\alpha$ , and the mode gate), including the two-line integration pattern for scikit-learn-style training loops. Experiments require only NumPy and scikit-learn (no GPU), consistent with the scope stated in Limitation L4; total runtime for the full table suite is under one CPU-hour.

## 10 Conclusion

K-ABENA v3 shows that threshold-based selective backpropagation and unbiased gradient estimation are compatible: a defensive-mixture design with inverse-probability weighting preserves the exact compute saving  $(1 - N)k/n$  while restoring the estimator guarantees that make SGD theory applicable (Lemmas 2–4, Theorem 1). The negative result (Proposition 2) delimits precisely why the uncompensated shortcut — shared by OHEM, SBP, and our own earlier variants — cannot converge when selection bias dominates the signal, and the experiments quantify both the failure (AUC 0.53 at 0.17% imbalance; accuracy 0.386 at 40% noise for the biased mode) and its resolution (0.9991; 0.808) at unchanged cost. We release the implementation with both modes and the gating logic [3]; the companion volume [2] develops the pedagogical treatment and is available upon request from the author.

## References

- [1] Yoshua Bengio, Jérôme Louradour, Ronan Collobert, and Jason Weston. Curriculum learning. In *Proceedings of the 26th Annual International Conference on Machine Learning (ICML)*, pages 41–48. ACM, 2009.
- [2] Jean-François Bonbhel. *The Art of Ignoring: K-ABENA and the Science of Selective Backpropagation — A Complete Theory of Error Thresholding and Proportional Exclusion in Machine Learning*. YekoElite University Press, Brazzaville, Republic of Congo, 2026. English edition. Available upon request from the author.
- [3] Jean-François Bonbhel. kabena-ml: K-abena python library (scikit-learn / pytorch / tensorflow). <https://github.com/Bonbhel/kabena-ml>, 2026. MIT License. Includes both estimator modes, validation scripts, and tutorials.
- [4] Saeed Ghadimi and Guanghui Lan. Stochastic first- and zeroth-order methods for nonconvex stochastic programming. *SIAM Journal on Optimization*, 23(4):2341–2368, 2013.
- [5] Daniel G. Horvitz and Donovan J. Thompson. A generalization of sampling without replacement from a finite universe. *Journal of the American Statistical Association*, 47(260):663–685, 1952.
- [6] Angela H Jiang, Daniel L-K Wong, Giulio Yang, Daniel Meng, Minjia Yu, Wei Xu, Cyrus Noghabi, Greg R Ganger, and Hai Helen Ke. Accelerating deep learning by focusing on the biggest losers. In *arXiv preprint arXiv:1910.00762*, 2019.
- [7] Angelos Katharopoulos and François Fleuret. Not all samples are created equal: Deep learning with importance sampling. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, 2018.
- [8] M. Pawan Kumar, Benjamin Packer, and Daphne Koller. Self-paced learning for latent variable models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 23, 2010.

- [9] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2980–2988, 2017.
- [10] Art B. Owen. Monte carlo theory, methods and examples. In *Available at statweb.stanford.edu/owen/mc*, 2013.
- [11] Carl-Erik Särndal, Bengt Swensson, and Jan Wretman. *Model Assisted Survey Sampling*. Springer, 1992.
- [12] Abhinav Shrivastava, Abhinav Gupta, and Ross Girshick. Training region-based object detectors with online hard example mining. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 761–769, 2016.

## A Proof of Lemma 3 (self-normalization bias)

Write  $\hat{T} = \sum_{i \notin \mathcal{M}} g_i + \sum_{i \in \mathcal{S}} \pi_i^{-1} g_i$  and  $\widehat{W} = (n - k) + \sum_{i \in \mathcal{S}} \pi_i^{-1}$ , so  $\hat{g} = \hat{T}/\widehat{W}$ ,  $\mathbb{E}[\hat{T}] = n\nabla F$ ,  $\mathbb{E}[\widehat{W}] = n$  (both by the indicator computation of Lemma 2). Using the identity

$$\frac{\hat{T}}{\widehat{W}} - \frac{\mathbb{E}[\hat{T}]}{n} = \frac{\hat{T} - \mathbb{E}[\hat{T}]}{n} + \hat{T} \left( \frac{1}{\widehat{W}} - \frac{1}{n} \right) = \frac{\hat{T} - \mathbb{E}[\hat{T}]}{n} - \frac{\hat{g}(\widehat{W} - n)}{n},$$

take expectations; the first term vanishes, giving the exact representation

$$\mathbb{E}[\hat{g}] - \nabla F = -\frac{1}{n} \mathbb{E}[\hat{g}(\widehat{W} - n)] = -\frac{1}{n} \text{Cov}(\hat{g}, \widehat{W}).$$

**First-order bound.** Since  $\hat{g}$  is a convex combination of per-sample gradients,  $\|\hat{g}\| \leq G_M$  almost surely; Cauchy–Schwarz on the covariance gives

$$\|\mathbb{E}[\hat{g}] - \nabla F\| \leq \frac{G_M}{n} \sqrt{\text{Var}(\widehat{W})}.$$

Under Assumption 1(b),  $\text{Var}(\widehat{W}) = \sum_{i \in \mathcal{M}} \frac{1 - \pi_i}{\pi_i} \leq \frac{k}{\alpha m/k} = \frac{k^2}{\alpha m}$  by Lemma 1, hence  $\|\mathbb{E}[\hat{g}] - \nabla F\| \leq G_M k / (n\sqrt{\alpha m})$  — the first bound of Lemma 3, of order  $O(1/\sqrt{m})$ .

**Second-order rate.** The Cauchy–Schwarz step is loose because  $\hat{g}$  concentrates: the standard ratio-estimator expansion [11, Ch. 5],

$$\mathbb{E}\left[\frac{\hat{T}}{\widehat{W}}\right] = \frac{\mathbb{E}[\hat{T}]}{\mathbb{E}[\widehat{W}]} - \frac{\text{Cov}(\hat{T}, \widehat{W})}{n^2} + \frac{\mathbb{E}[\hat{T}] \text{Var}(\widehat{W})}{n^3} + o(1/m),$$

has both correction terms bounded by  $G_M \text{Var}(\widehat{W})/n^2$  in norm (using  $\|\text{Cov}(\hat{T}, \widehat{W})\| \leq G_M \text{Var}(\widehat{W})$ , since each coordinate of  $\hat{T} - \mathbb{E}\hat{T}$  is a weight-linear combination of gradient coordinates bounded by  $G_M$  times the corresponding fluctuation of  $\widehat{W}$ , and  $\|\mathbb{E}[\hat{T}]\|/n \leq G_M$ ). Therefore

$$\delta_m \leq \frac{2 G_M \text{Var}(\widehat{W})}{n^2} + o(1/m) \leq \frac{2 G_M k^2}{\alpha m n^2} + o(1/m) = O(1/m).$$

At  $\alpha = 1$  the design is SRSWOR with  $\pi_i = m/k$  exactly,  $\text{Var}(\widehat{W}) = \frac{k^2}{m} \cdot \frac{k-m}{k} \cdot \frac{k-m}{k-1} \leq \frac{k(k-m)}{m}$ , giving  $\delta_m \leq 2G_M k(k-m)/(mn^2) + o(1/m)$ . Both regimes are  $O(1/m)$ ; the empirical values (0.004 at large  $n$ ; 0.037 in the adversarial micro-regime) sit inside these bounds.  $\square$

## B Proof of Theorem 1

Write  $b_t = \mathbb{E}[\hat{g}_t \mid \theta_t] - \nabla F(\theta_t)$  ( $\|b_t\| \leq \delta$ ) and  $\xi_t = \hat{g}_t - \mathbb{E}[\hat{g}_t \mid \theta_t]$  ( $\mathbb{E}[\xi_t \mid \theta_t] = 0$ ,  $\mathbb{E}\|\xi_t\|^2 \leq \hat{\sigma}^2$ ).  $L$ -smoothness gives

$$F(\theta_{t+1}) \leq F(\theta_t) - \eta \langle \nabla F(\theta_t), \hat{g}_t \rangle + \frac{L\eta^2}{2} \|\hat{g}_t\|^2.$$

Take conditional expectations; using  $\mathbb{E}[\hat{g}_t \mid \theta_t] = \nabla F(\theta_t) + b_t$  and  $\mathbb{E}\|\hat{g}_t\|^2 \leq \|\nabla F(\theta_t) + b_t\|^2 + \hat{\sigma}^2 \leq 2\|\nabla F(\theta_t)\|^2 + 2\delta^2 + \hat{\sigma}^2$ :

$$\mathbb{E}[F(\theta_{t+1})] \leq \mathbb{E}[F(\theta_t)] - \eta \mathbb{E}\|\nabla F(\theta_t)\|^2 - \eta \mathbb{E}\langle \nabla F(\theta_t), b_t \rangle + \frac{L\eta^2}{2} (2\mathbb{E}\|\nabla F(\theta_t)\|^2 + 2\delta^2 + \hat{\sigma}^2).$$

Bound  $-\mathbb{E}\langle \nabla F, b_t \rangle \leq \delta \mathbb{E}\|\nabla F(\theta_t)\| \leq \delta G_\infty$ . With  $\eta \leq \frac{1}{2L}$ ,  $L\eta^2 \leq \eta/2$ , so the  $\mathbb{E}\|\nabla F\|^2$  terms combine into  $-\frac{\eta}{2} \mathbb{E}\|\nabla F(\theta_t)\|^2$  at worst. Rearranging and summing over  $t = 0, \dots, T-1$ , then dividing by  $\eta T/2$ :

$$\min_t \mathbb{E}\|\nabla F(\theta_t)\|^2 \leq \frac{1}{T} \sum_t \mathbb{E}\|\nabla F(\theta_t)\|^2 \leq \frac{2(F(\theta_0) - F^*)}{\eta T} + L\eta(\hat{\sigma}^2 + 2\delta^2) + 2\delta G_\infty.$$

Substituting  $\eta = \min\{\frac{1}{2L}, \frac{c}{\sqrt{T}}\}$  yields the statement (absorbing constants; the  $+\delta^2$  of the main text majorizes  $2\delta^2 \cdot Lc/\sqrt{T}$  for  $T$  large). With  $\delta = 0$  (HT form) the bound is the classical rate of Ghadimi and Lan [4].  $\square$

## C Ablation history

The canonical design is the outcome of a systematic exploration; all rejected variants are documented to preempt re-exploration. Reference: v1,  $p_i \propto (K - \ell_i)$  over all of  $\mathcal{M}$ , no reweighting. **H1** ( $p_i \propto \ell_i$  over all of  $\mathcal{M}$ , no reweighting): bias +18 to +26%, variance +29% vs. v1 — rejected; note that this same proposal distribution becomes near-optimal *once inverse-probability weighting is added*, which is precisely the v3 construction. **H2** ( $p_i \propto |\ell_i - c|$ , symmetric U): interpolates v1 and H1, bias +6 to +12% — rejected. **H4** (upper half only,  $p_i \propto K - \ell_i$  on  $\mathcal{U}$ ): bias +60 to +80% — rejected; both terms of Lemma 5 then share a sign. **v2** (lower half only,  $p_i \propto \ell_i$  on  $\mathcal{L}$ ): bias -4 to -10% and variance -25 to -55% vs. v1, weight ratio bounded ( $r \approx 1.4$  vs.  $r \approx 3 \times 10^4$  for v1; 0/40 vs. 14/40 violations of the sampling approximation at 1% tolerance) — adopted as an interim canonical form before the end-to-end extreme-imbalance and label-noise tests of Tables 2–4 revealed the shared failure of all uncompensated variants and motivated the compensated design. Equivalence sweeps: at matched bias, v2 allows no additional retention saving at  $N = 0.3$  and +3.3% at  $N = 0.4$  relative to v1 — the value of the uncompensated refinements was always estimator quality at fixed cost, never additional cost reduction, a conclusion that transfers to v3.